

1

DOCUMENT REPRESENTATION

Scientific PDF



Layout-aware parsing



Evidence objects

Text

Figure

Table

page + bounding box + source ID

2

RETRIEVAL + LOCAL ANSWERING

User question



BM25

Dense

Evidence selection

RRF

Local model

Cited draft

[1] [2] [...]

3

VERIFICATION + INSPECTION

Deterministic checks

lexical + numerical



Final cited answer

PDF region
highlightEvidence
Graph
trace

click citation → source region

Source identity preserved across the pipeline

document ID • page • bounding box • evidence ID